Making Better Decisions with Data

A Primer on Marketing and Business Analytics

Blindspot Lab Demonstration Unit · Chapter Pack

About this unit

This short primer introduces six connected ideas that marketing and business analysts use to make decisions: revenue and conversion metrics, contribution margin, return on advertising spend, customer acquisition cost and lifetime value, correlation versus causation, and decision-making under changing assumptions. Each chapter builds on the previous ones.

Synthetic teaching material created for the Blindspot Lab demonstration. All companies, prices and results are fictional.

Chapter 1. Revenue and Conversion Metrics

1.1 Revenue

Revenue is the money a business takes in from sales before any costs are deducted. For a single product, revenue equals the selling price multiplied by the number of units sold. Revenue is a measure of scale, not of profit: a business can grow revenue while losing money on every sale.

Revenue = selling price × units sold

1.2 Conversion rate and absolute conversions

The conversion rate is the share of visitors (or leads) who complete a desired action such as a purchase. A high conversion rate does not guarantee a large number of conversions. A landing page that converts 8% of 500 visitors produces 40 customers, while a page that converts 3% of 4,000 visitors produces 120 customers. Decisions about where to invest should consider absolute conversions and the value of each conversion, not only the rate.

Conversion rate = conversions ÷ visitors

Table 1.1 Synthetic funnel data for Northwind Outdoor Co. (fictional), one month

Channel	Visitors	Conversion rate	Conversions	Avg order value	Revenue
Email	5,000	4.0%	200	₹2,500	₹500,000
Paid search	20,000	2.5%	500	₹2,200	₹1,100,000
Social	40,000	0.8%	320	₹1,800	₹576,000
Affiliates	8,000	3.0%	240	₹2,000	₹480,000

In Table 1.1, email has the highest conversion rate, but paid search produces the most conversions and the most revenue because it reaches many more visitors.

1.3 Percentage points versus percent change

When a rate changes, there are two different ways to describe the change. A percentage-point change is the simple difference between two percentages: a conversion rate that rises from 2% to 3% has increased by 1 percentage point. A percent change is the difference relative to the starting value: the same move from 2% to 3% is a 50% increase. Reports should state which measure they use, because confusing the two can make a small change look large or a large change look small.

Chapter 2. Contribution Margin

2.1 Variable costs and fixed costs

Variable costs change in proportion to the number of units sold, for example materials, packaging, payment fees and delivery. Fixed costs, such as rent and salaried staff, do not change with short-run volume. Separating the two is the first step in understanding how a decision affects profit.

2.2 Unit contribution and the contribution margin ratio

Contribution margin per unit is the selling price minus the variable cost per unit. It is the amount each additional unit contributes towards covering fixed costs and then generating profit. Total contribution equals the contribution margin per unit multiplied by the number of units sold. The contribution margin ratio expresses contribution as a share of the selling price.

> **Contribution margin per unit = price − variable cost per unit**
>
> **Contribution margin ratio = (price − variable cost) ÷ price**

Example: a product sells for ₹500 and has a variable cost of ₹300 per unit. Its contribution margin per unit is ₹200 and its contribution margin ratio is 40%.

2.3 Price changes and the volume needed to compensate

A price cut reduces the contribution of every unit sold, not just the revenue. Because variable costs stay the same, the contribution per unit falls by the full amount of the price cut. To keep total contribution unchanged, volume must rise by the ratio of the old to the new contribution per unit.

> **Required volume increase = (old contribution per unit ÷ new contribution per unit) − 1**

Continuing the example, cutting the price from ₹500 to ₹450 lowers contribution per unit from ₹200 to ₹150. Volume must rise by 200 ÷ 150 − 1 = 33.3% just to earn the same total contribution. Keeping revenue constant would need only an 11.1% increase (500 ÷ 450 − 1), which is why plans that aim to 'make up the revenue' can still leave the business with less contribution to cover its fixed costs.

Preserving revenue and preserving contribution are different goals. Revenue preservation ignores variable costs; contribution preservation accounts for them.

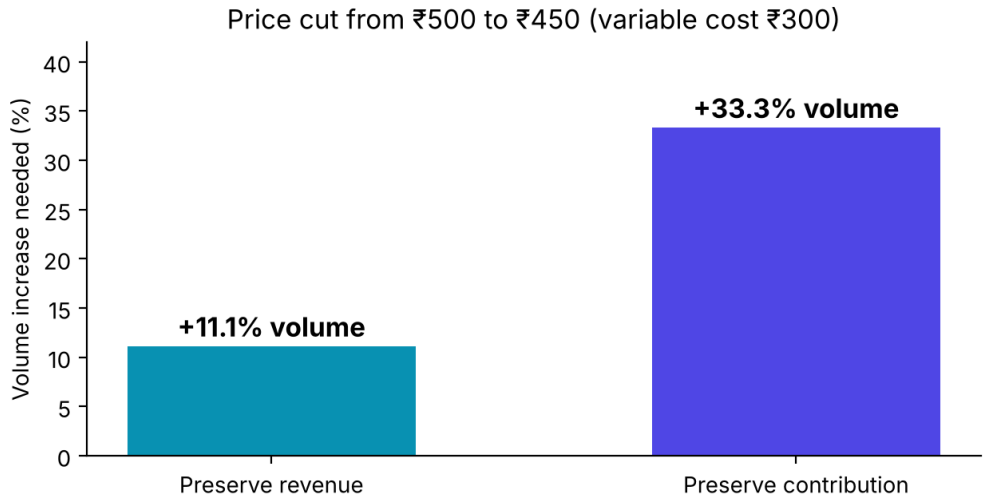


Figure 2.1 Volume increase required after a 10% price cut (synthetic example). Preserving contribution requires three times the volume increase needed to preserve revenue.

2.4 Total cost versus marginal cost

For decisions about one additional order, what matters is the marginal (incremental) cost of serving it, usually the variable cost, not the average total cost per unit, which includes fixed costs that will be paid anyway. Example: a workshop with a variable cost of ₹300 and an average total cost of ₹380 per unit (including allocated rent) receives a one-off order for 100 units at ₹340 each. If spare capacity exists and fixed costs will not change, the order adds (340 − 300) × 100 = ₹4,000 of contribution, even though ₹340 is below the average total cost.

2.5 Break-even volume

Break-even volume is the number of units at which total contribution exactly covers fixed costs. With fixed costs of ₹60,000 and contribution per unit of ₹200, the business must sell 300 units to break even.

Break-even volume = fixed costs ÷ contribution margin per unit

Chapter 3. Return on Advertising Spend

3.1 Definition

Return on advertising spend (ROAS) is the revenue attributed to advertising divided by the advertising cost. A campaign that spends ₹50,000 and is credited with ₹200,000 of revenue has a ROAS of 4, often written 4x or 400%.

ROAS = attributed revenue ÷ advertising spend

3.2 ROAS is not profitability

ROAS compares advertising cost with revenue, not with profit. Because revenue still has to pay for the variable costs of the products sold, a ROAS above 1 does not mean the campaign made money. The break-even ROAS is the ROAS at which the contribution from attributed sales exactly covers the advertising spend. A business with a 40% contribution margin ratio needs a ROAS of at least 2.5 before advertising contributes anything towards fixed costs.

Break-even ROAS = 1 ÷ contribution margin ratio

Contribution after advertising = attributed revenue × contribution margin ratio − advertising spend

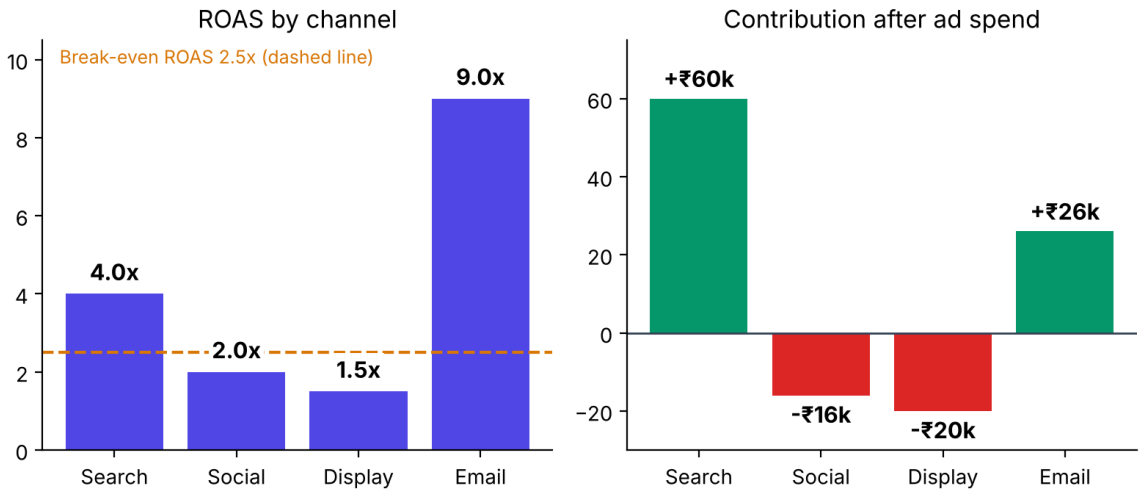


Figure 3.1 ROAS and contribution after advertising by channel for Northwind Outdoor Co. (synthetic data, contribution margin ratio 40%, break-even ROAS 2.5).

3.3 Attribution and incrementality

ROAS depends on how revenue is attributed to campaigns. Some attributed sales would have happened without the advertising, so ROAS can overstate the incremental effect of a campaign. Incrementality tests, which compare exposed and unexposed groups, give a better estimate of the causal effect.

Chapter 4. Customer Acquisition Cost and Lifetime Value

4.1 Customer acquisition cost

Customer acquisition cost (CAC) is the total sales and marketing spend in a period divided by the number of new customers acquired in that period. Spending ₹300,000 to acquire 1,200 new customers gives a CAC of ₹250.

> **CAC = sales and marketing spend ÷ new customers acquired**

4.2 Customer lifetime value

Customer lifetime value (CLV) estimates the total contribution a customer generates over their relationship with the business. A simple version multiplies the average monthly contribution per customer by the expected customer lifetime in months. Using revenue instead of contribution overstates CLV, because it ignores the cost of serving the customer. For example, a customer who pays ₹500 per month at a 40% contribution margin ratio generates ₹200 of contribution per month; with an expected lifetime of 18 months, CLV is ₹3,600.

> **CLV = average monthly contribution per customer × expected lifetime in months**

4.3 Comparing CAC with CLV

A CLV-to-CAC ratio compares the value a customer generates with the cost of acquiring them. Many analysts look for a ratio comfortably above 1, often around 3, but the right threshold depends on the business. The payback period is the number of months of contribution needed to recover CAC: with a CAC of ₹900 and monthly contribution of ₹200, payback takes 4.5 months.

> **Payback period (months) = CAC ÷ monthly contribution per customer**

A low CAC is not automatically good. Cheap customers who churn quickly can be worth less than more expensive customers who stay for years. Always compare CAC with contribution-based lifetime value.

Chapter 5. Correlation, Causation and Experiments

5.1 Correlation is not causation

Two variables are correlated when they tend to move together. Correlation alone does not show that one causes the other. Customers who use a loyalty app may spend more than other customers, but loyal, high-spending customers may simply be more likely to install the app in the first place. A third variable that influences both, called a confounder, can create a correlation without any causal link.

5.2 Randomised experiments

A randomised controlled experiment, often called an A/B test, assigns customers to groups at random so that confounders are balanced on average. If the treated group then behaves differently from the control group, the difference can be attributed to the treatment, within the limits of sampling error.

5.3 Statistical significance versus practical significance

A statistically significant result is one that would be unlikely if there were truly no effect; it says little about whether the effect is large enough to matter. With very large samples, a 0.1 percentage-point lift in conversion can be statistically significant yet too small to justify the cost of a change. Practical significance asks whether the size of the effect is worth acting on, considering costs, risks and the confidence interval.

Chapter 6. Decisions under Changing Assumptions

6.1 Sensitivity and scenario analysis

Business cases depend on assumptions such as price, variable cost, conversion rate and churn. Sensitivity analysis changes one assumption at a time to see how much the result moves. Scenario analysis changes several assumptions together to describe a pessimistic, expected and optimistic case.

Table 6.1 Campaign scenarios for 50,000 visitors and ₹100,000 of advertising spend (synthetic)

Assumption	Pessimistic	Expected	Optimistic
Conversion rate	1.5%	2.0%	2.5%
Average order value	₹1,800	₹2,000	₹2,200
Contribution margin ratio	35%	40%	45%
Contribution after ad spend	₹372,500	₹700,000	₹1,137,500

6.2 Mean versus median

Averages can mislead when data are skewed. Order values often contain a few very large orders. For the order values ₹800, ₹900, ₹1,000, ₹1,100 and ₹9,200, the mean is ₹2,600 but the median is ₹1,000. The median better describes a typical order, while the mean is pulled up by the single large order. Use the median to describe a typical customer and the mean when you need totals, because the mean multiplied by the count gives the total.

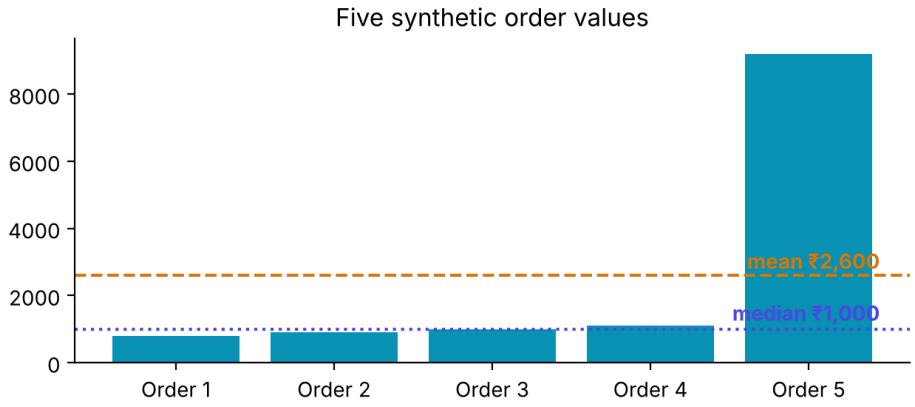


Figure 6.1 One very large order pulls the mean far above the median (synthetic data).

6.3 A decision checklist

First, state the decision and the metric that matters, which is usually contribution rather than revenue. Second, separate correlation from causal evidence. Third, test how sensitive the conclusion is to the key assumptions. Fourth, prefer incremental measures over attributed or average measures.

Glossary

Break-even ROAS: the ROAS at which contribution from attributed sales equals advertising spend; 1 ÷ contribution margin ratio.

Contribution margin: selling price minus variable cost, per unit or in total.

Confounder: a variable that influences both a supposed cause and its effect.

Conversion rate: conversions divided by visitors or leads.

Customer acquisition cost (CAC): sales and marketing spend divided by new customers acquired.

Customer lifetime value (CLV): expected total contribution from a customer relationship.

Percentage point: the arithmetic difference between two percentages.

Variable cost: a cost that rises in proportion to units sold.

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